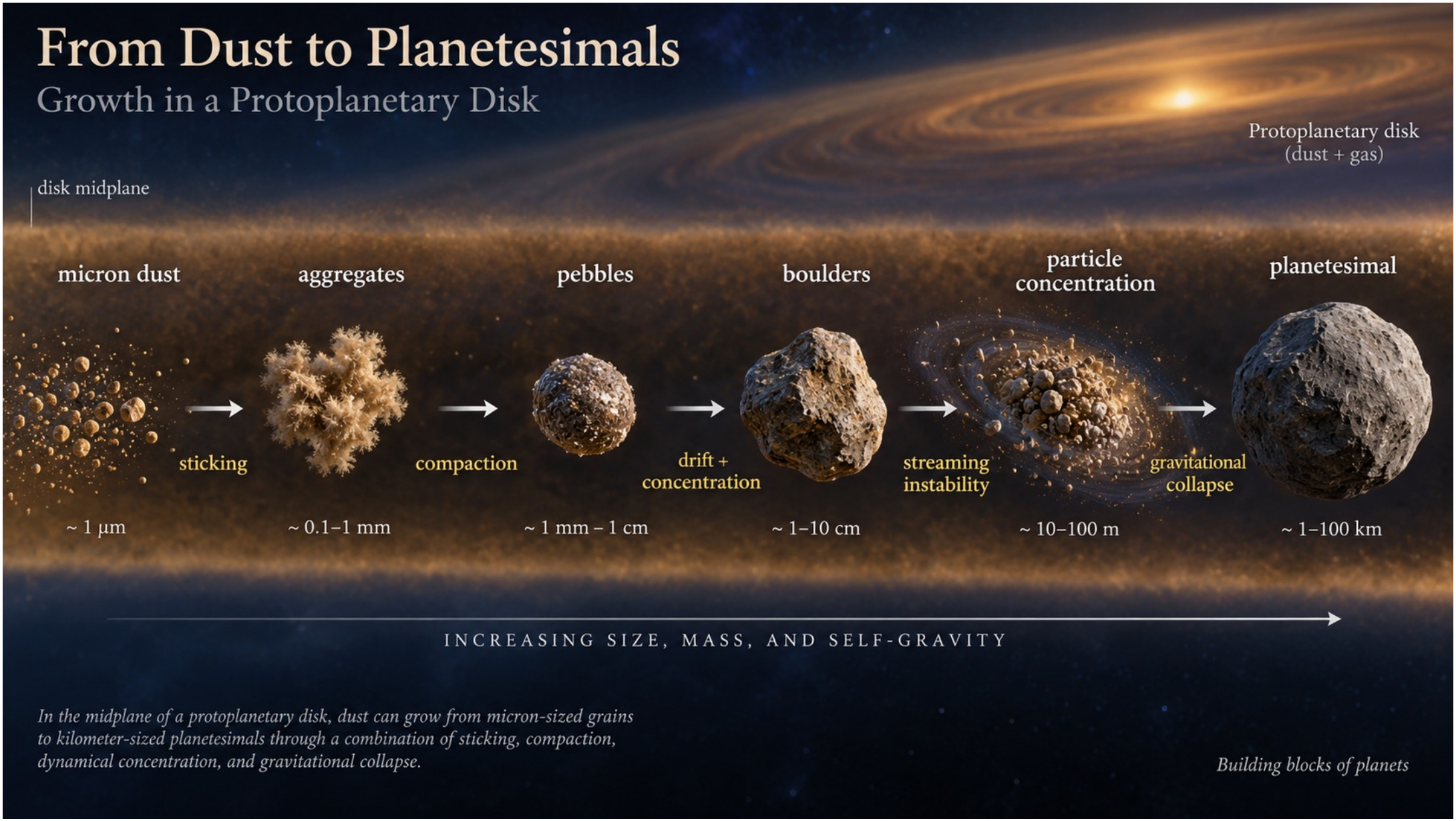

From Dust to Planetesimals
Growth in a Protoplanetary Disk
Protoplanetary disk
(dust + gas)
disk midplane
micron dust
aggregates
pebbles
boulders
particle concentration
planetesimal
sticking
compaction
drift + concentration
streaming instability
gravitational collapse
~ 1 μm
~ 0.1–1 mm
~ 1 mm – 1 cm
~ 1–10 cm
~ 10–100 m
~ 1–100 km
INCREASING SIZE, MASS, AND SELF-GRAVITY
In the midplane of a protoplanetary disk, dust can grow from micron-sized grains to kilometer-sized planetesimals through a combination of sticking, compaction, dynamical concentration, and gravitational collapse.
Building blocks of planets

Gas–Dust Dynamics in a Protoplanetary Disk

FROM MICRON GRAINS TO PLANETS

YOUNG SYSTEM
BRIGHT FUTURES

Pressure-supported gas

(orbits slightly sub-Keplerian)

Dust feels a headwind

Young star

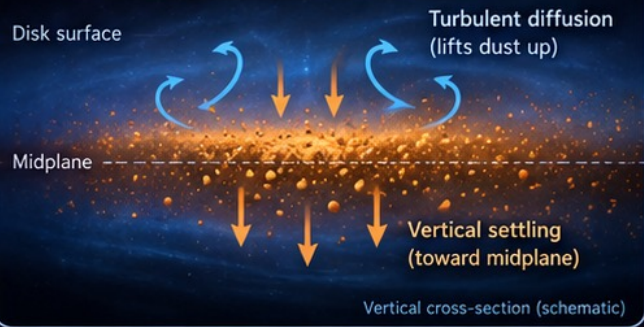
Radial drift

Backreaction
Dust can modify the gas motion

Dust concentration / pile-up



Vertical settling vs. turbulent diffusion



Gas (H, He + trace species)

Dust (μm – km)

Stokes number controls coupling



BUILDING PLANETS FROM DUST

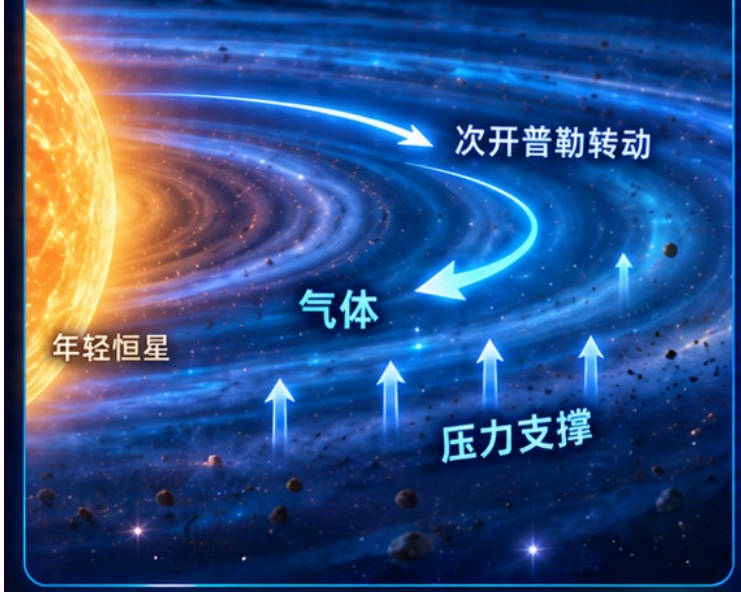
PLANET FORMATION IN ACTION

原行星盘中的径向漂移

为什么尘埃会向恒星漂移？

1 气体

气体受压强支撑，转动略慢于开普勒速度



2 尘埃

尘埃本想按开普勒速度运动
但受到气体的迎风阻力



3 结果

尘埃被拖慢，逐渐向恒星靠近



关键结论： 压强先改变气体转动；尘埃再通过拖曳力间接受到影响。

行星从星尘中诞生

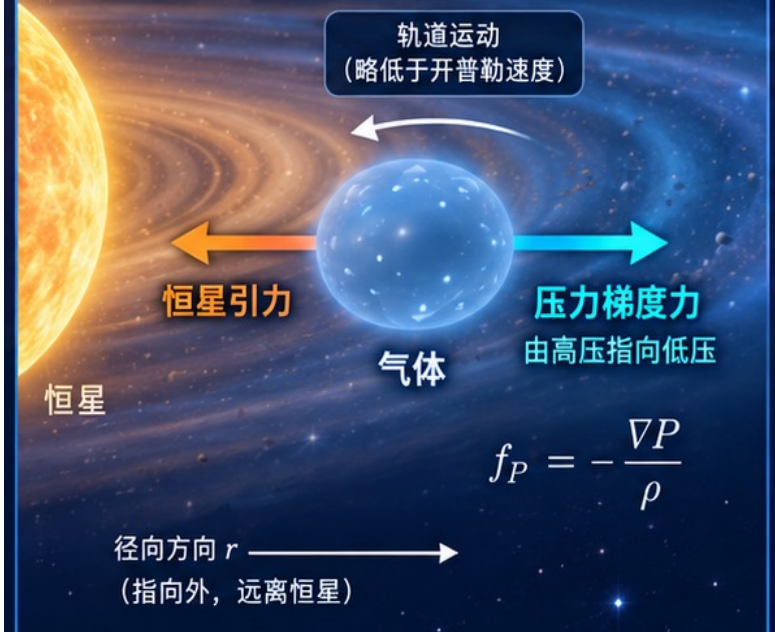
原行星盘中气体与尘埃的受力

—— 为什么尘埃不直接感受到气体的压力梯度？

From dust to planets in protoplanetary disks

气体受力

气体是连续流体，直接受到压力梯度力



$$f_P = -\frac{\nabla P}{\rho}$$

气体是连续流体，直接受到压力梯度力。

尘埃受力

尘埃是离散固体颗粒



尘埃是离散固体颗粒。

为什么尘埃没有直接感受到 pressure?

1. 压力是气体流体的宏观量，力密度为 $-\nabla P$。



2. 单个尘埃粒子不属于气体连续介质，因此没有显著的直接压力梯度力。

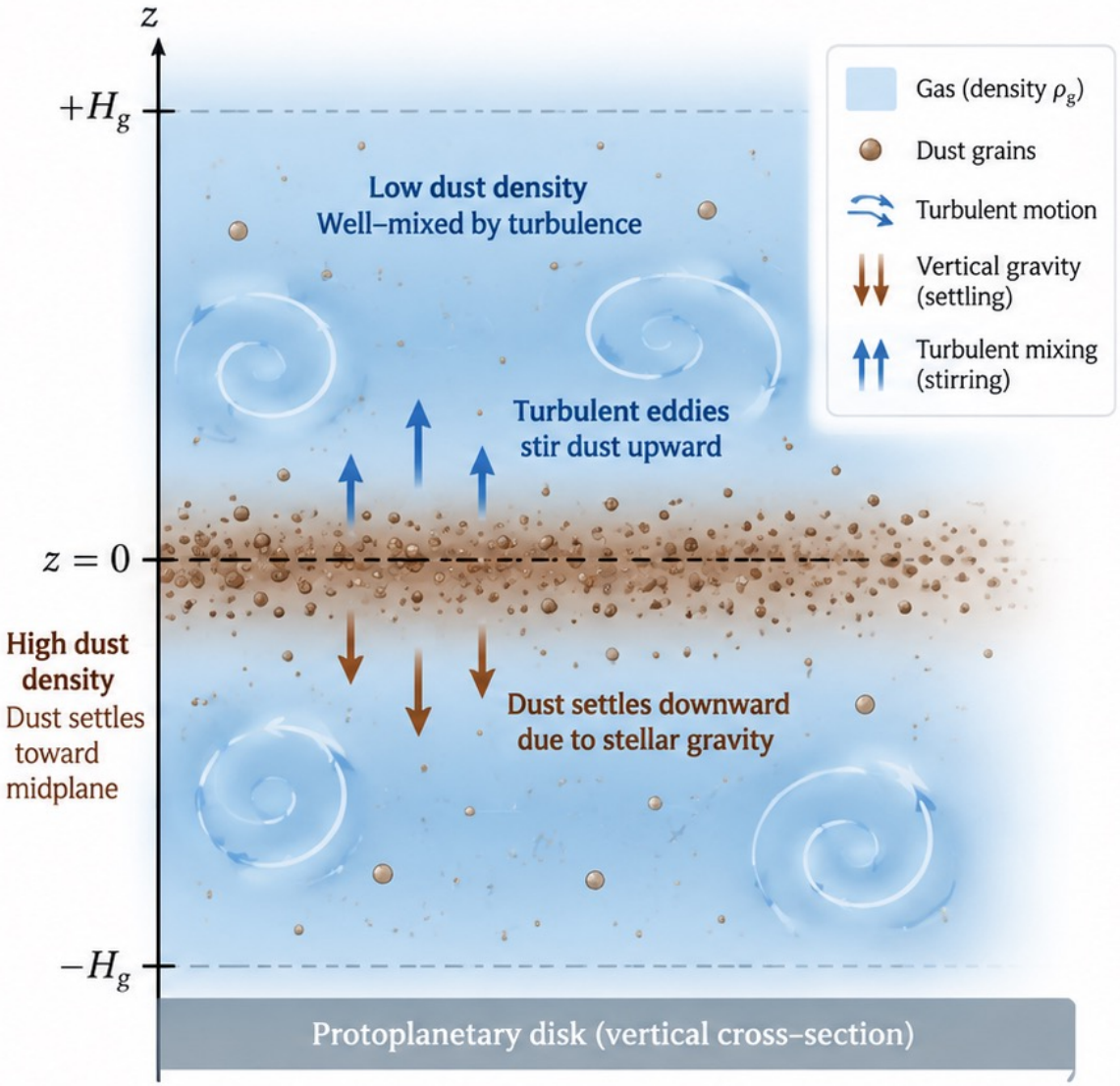


3. 尘埃与气体的相互作用主要来自表面分子碰撞，因此表现为拖曳力 drag force。



结论： 气体直接受 pressure gradient force；尘埃只通过 gas drag 间接受到气体影响。

Vertical settling vs. turbulent stirring of dust



Competition between settling and turbulent diffusion

Gravitational settling

$$v_{\rm set} \approx -\Omega_K^2\, t_{\rm stop}\, z \quad \text{(for small St)}$$

Pulls dust toward the midplane

Turbulent diffusion

$$F_{\rm diff} = -D_{\rm d}\,\frac{\partial \rho_{\rm d}}{\partial z}$$

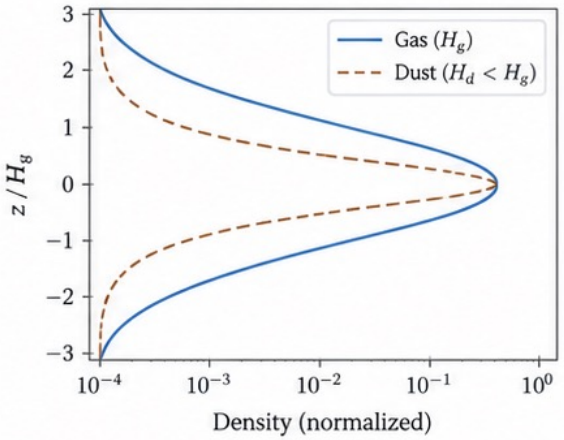
Turbulence mixes dust upward

Equilibrium (settling–stirring balance)

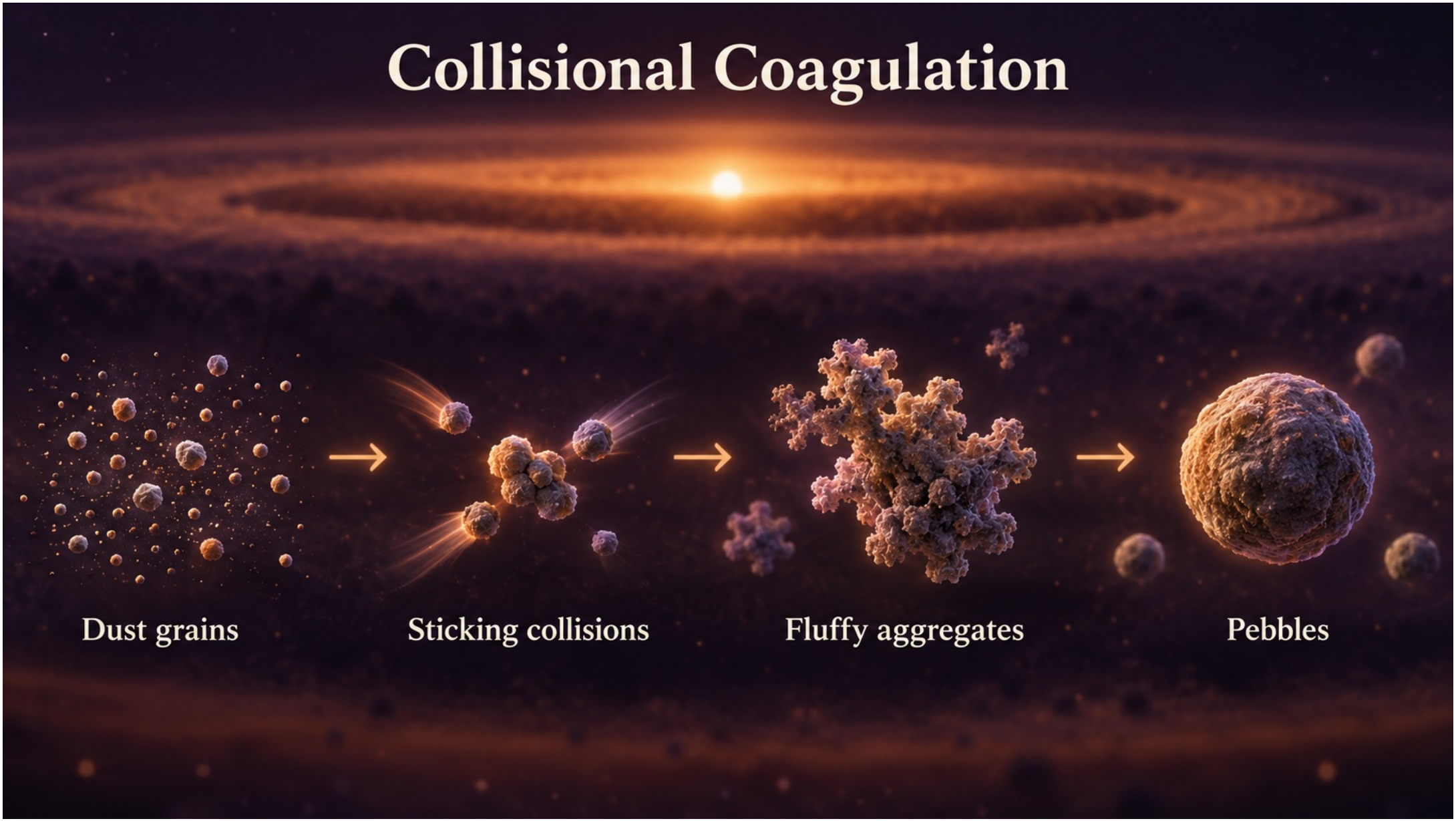
Vertical dust distribution

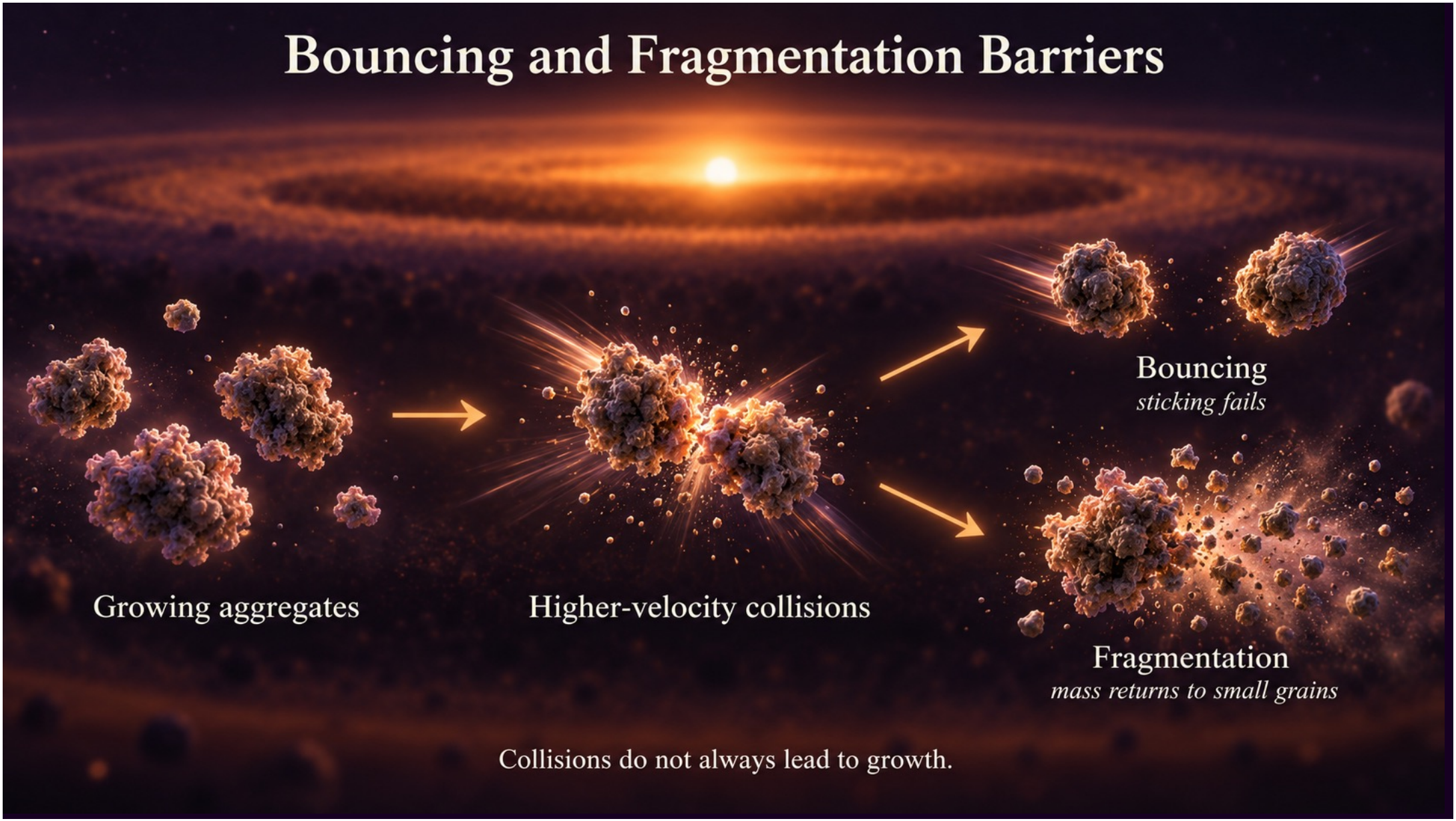
$$\rho_d(z) = \rho_{\rm d,0}\exp\left(-\frac{z^2}{2H_{\rm d}^2}\right) \qquad H_{\rm d} = H_{\rm g}\sqrt{\frac{\alpha}{\alpha+{\rm St}}}$$

Vertical profiles (illustration)



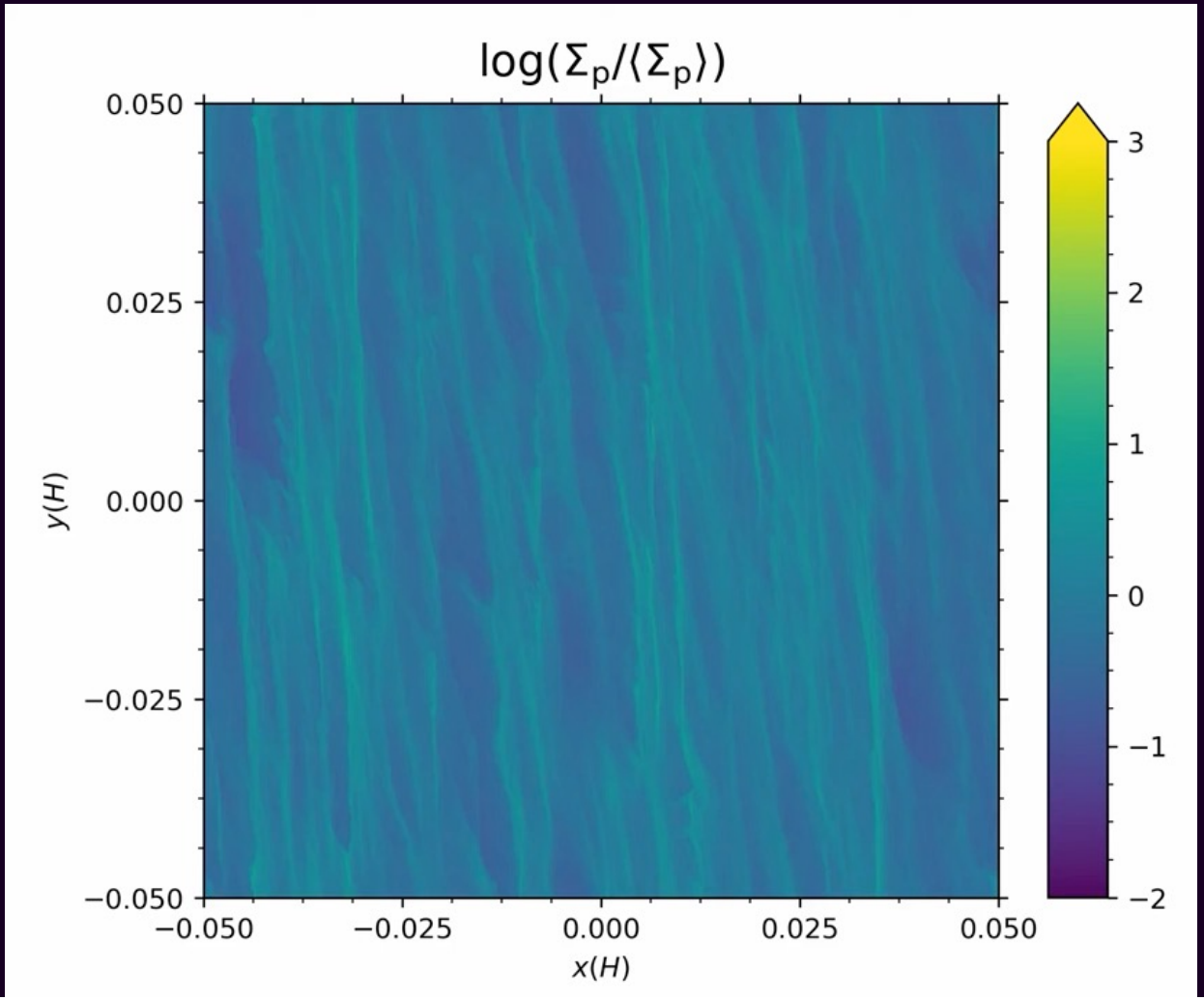
- Gas is supported by pressure, so it has a large scale height $H_g = c_s/\Omega_K$.
- Dust feels little pressure force (because the pressure gradient acts on gas; dust feels it only indirectly via gas drag), and thus settles toward the midplane.
- Turbulence counteracts settling, leading to a finite dust scale height H_d.
- Larger St (bigger grains) → thinner dust layer.
- Stronger turbulence (larger α) → thicker dust layer.


Collisional Coagulation
Dust grains
Sticking collisions
Fluffy aggregates
Pebbles


Bouncing and Fragmentation Barriers
Growing aggregates
Higher-velocity collisions
Bouncing
sticking fails
Fragmentation
mass returns to small grains
Collisions do not always lead to growth.

流不稳定性

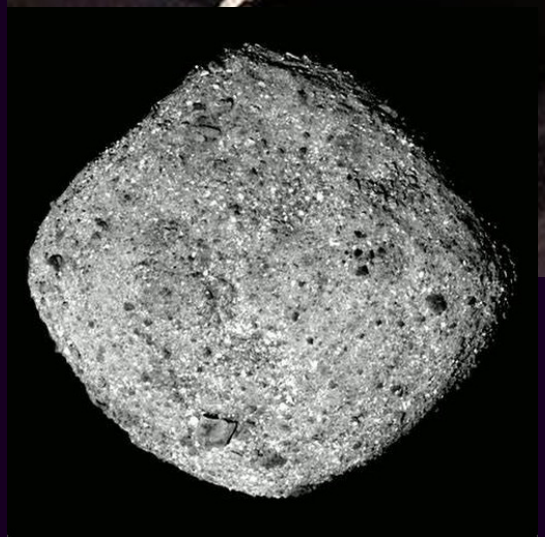
STREAMING INSTABILITY: NUMERICAL RESULTS

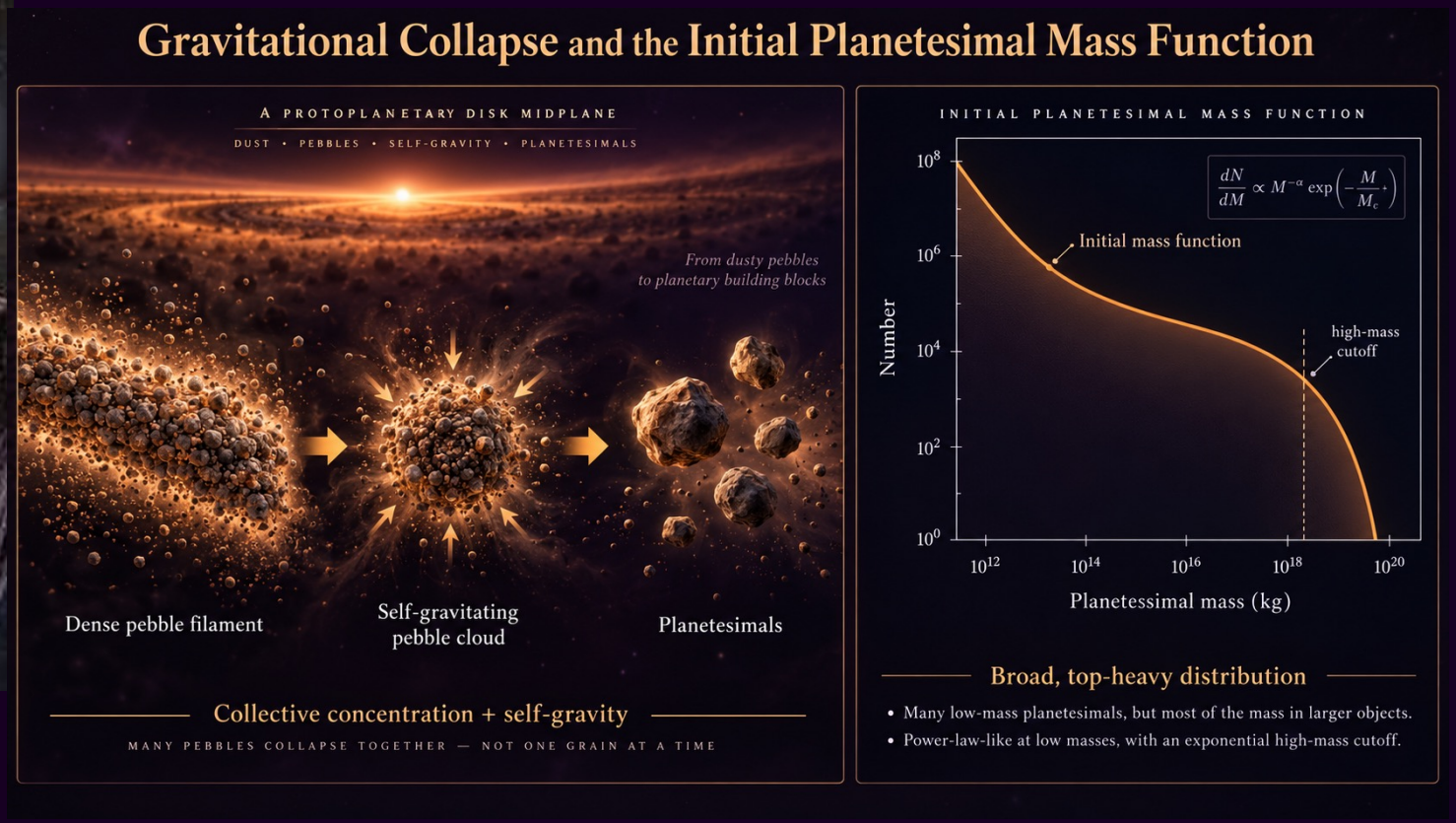


引力坍缩与初始星子质量函数

GRAVITATIONAL COLLAPSE AND THE INITIAL PLANETESIMAL MASS FUNCTION







Asteroid Belt and Kuiper Belt: Fossil Records of Planetesimal Formation

Not all planetesimals became planets — some survived, preserving the early history of our Solar System.

1 Protoplanetary disk
(dust and pebbles, ~ 0–1 Myr)



- Gas, dust and pebbles in a rotating disk around a young star.
- Initial building blocks of planetesimals.

2 Planetesimal formation
(~ 0.1–10 Myr)



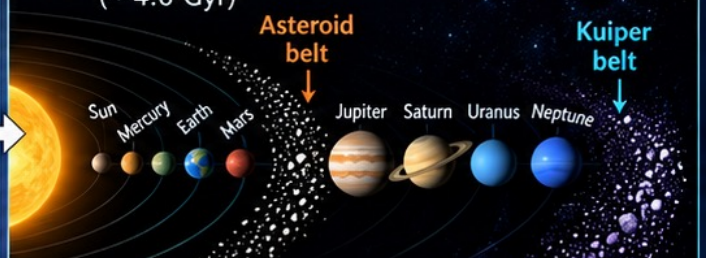
- Streaming instability and other processes concentrate pebbles.
- Gravitational collapse forms km–100 km planetesimals.

3 Growth and dynamical evolution
(~ 10–100 Myr)



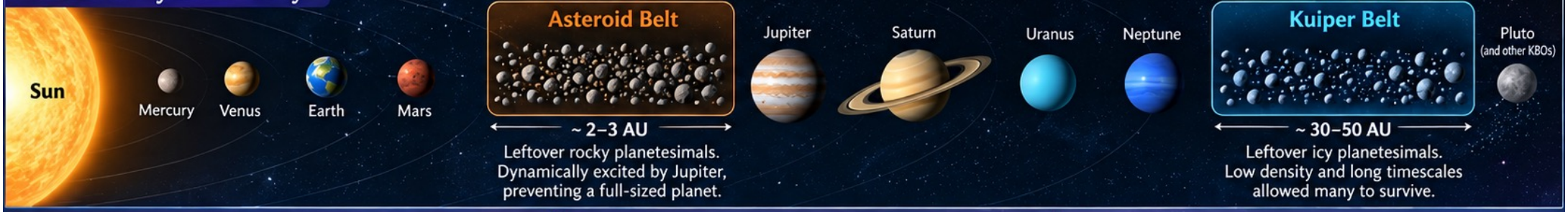
- Some planetesimals accrete into planetary embryos and planets.
- Others are scattered, excited, or remain small.

4 Present-day Solar System
(~ 4.6 Gyr)



- Leftover planetesimals survive as asteroids and Kuiper-belt objects.
- They are a fossil record of the early Solar System.

The Solar System Today



Leftover rocky planetesimals. Dynamically excited by Jupiter, preventing a full-sized planet.

Leftover icy planetesimals. Low density and long timescales allowed many to survive.

Examples of asteroid belt objects (mostly rocky)



Ceres	Vesta	Pallas	Hygiea
(~ 940 km)	(~ 525 km)	(~ 512 km)	(~ 430 km)

Examples of Kuiper belt objects (icy)



Pluto	Haumea	Makemake	Eris
(~ 2,377 km)	(~ 1,632 km)	(~ 1,430 km)	(~ 2,326 km)

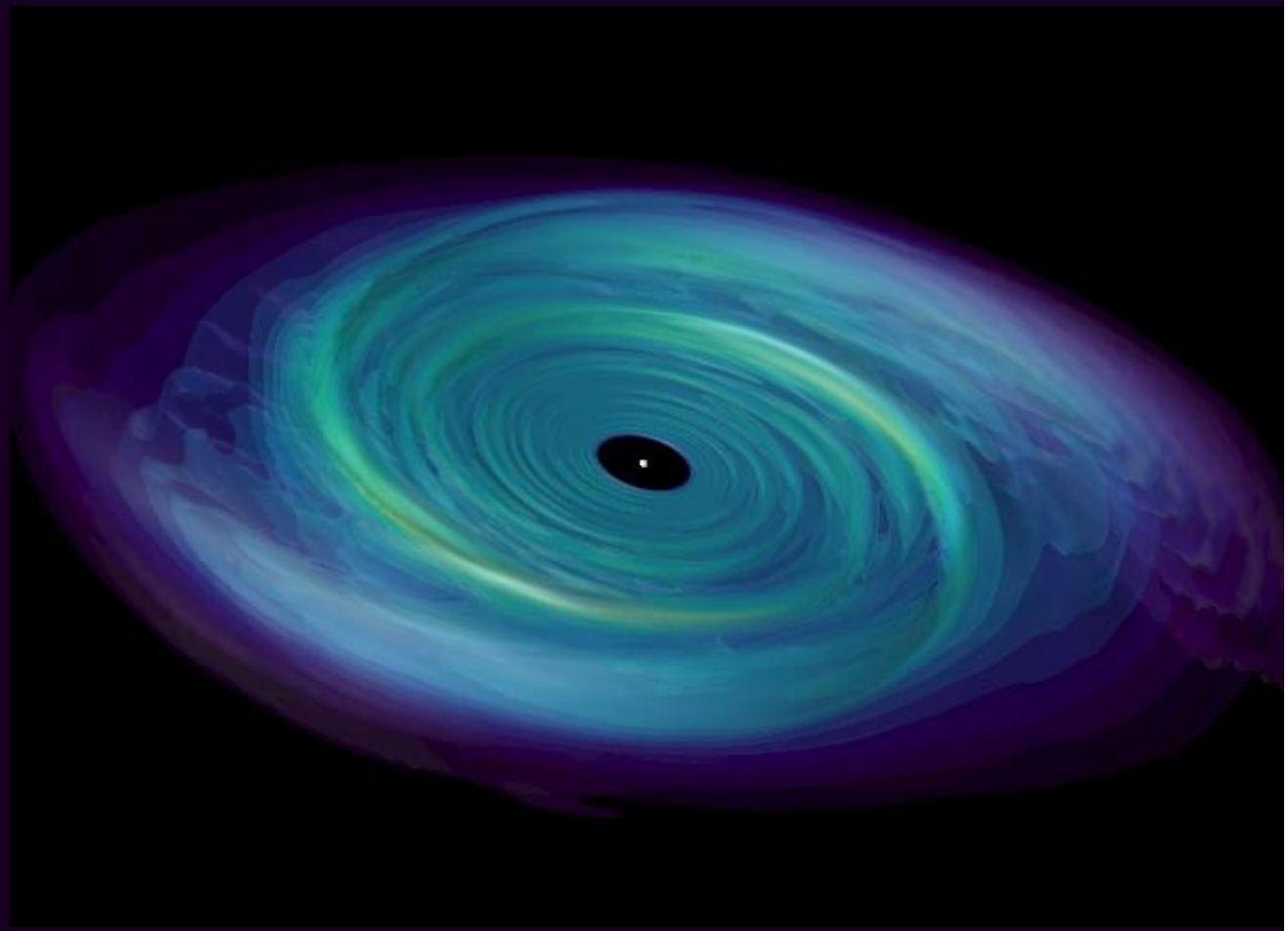
What they tell us

- Preserve the primordial composition of the protoplanetary disk.
- Record the size distribution of early planetesimals.
- Constrain models of planetesimal formation and planetary migration.
- Provide clues to the dynamical history of the Solar System.

Asteroid belt and Kuiper belt are time capsules — surviving planetesimals that did not get incorporated into planets.

尘埃动力学

Dust Dynamics



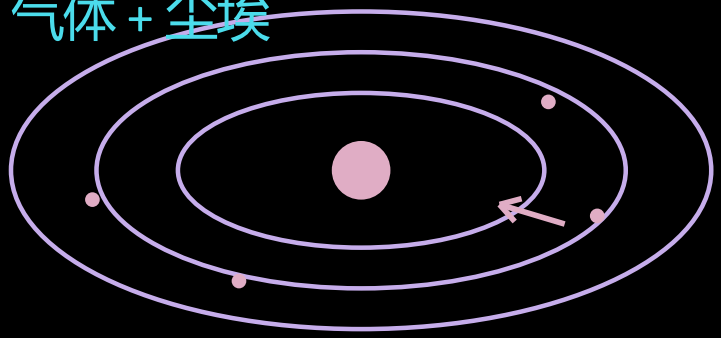
1 原行星盘中的尘埃动力学

DUST DYNAMICS IN PROTOPLANETARY DISKS

$$\text{gravity} + \text{drag} + \text{diffusion}$$

- 尘埃是形成星子与行星的固体原料
- 尘埃和气体承受不同的压强梯度力
- 耦合强度决定沉降、漂移与捕获

气体 + 尘埃



阻力连接气体与尘埃

2 气体与尘埃：受力为何不同?

FORCES ON GAS AND DUST

$$\frac{D_g \mathbf{v}_g}{Dt} = -\nabla\Phi - \frac{\nabla P}{\rho_g} + \frac{\rho_d}{\rho_g}\frac{\mathbf{v}_d - \mathbf{v}_g}{t_s}, \qquad \frac{D_d \mathbf{v}_d}{Dt} = -\nabla\Phi - \frac{\mathbf{v}_d - \mathbf{v}_g}{t_s}$$

- Φ：恒星引力势；P：气体压力；ρ_g：气体密度
- v_g、v_d：气体与尘埃速度；t_s：停止时间
- D_g/Dt、D_d/Dt 分别沿两种组分的运动求导
- 本页忽略黏性、磁力、尘埃反馈与自引力；尘埃视为无压流体

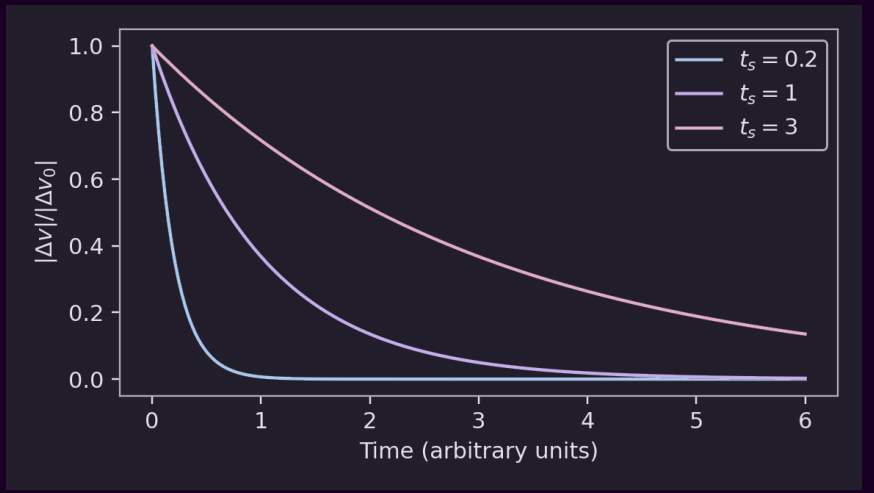
气体有压力支撑，尘埃无。尘埃/气体通过drag force感受对方。

3 停止时间：相对速度衰减的时间尺度

STOPPING TIME

$$\frac{d\Delta\mathbf{v}}{dt} = -\frac{\Delta\mathbf{v}}{t_s} \quad \Rightarrow \quad \Delta\mathbf{v}(t) = \Delta\mathbf{v}_0 e^{-t/t_s}$$

- $\Delta v = v_d - v_g$：气体与尘埃的相对速度
- t_s 越短，尘埃和气体耦合越强



4 Stokes 数：把阻力与轨道周期比较

THE STOKES NUMBER

$$\mathrm{St} = \Omega_K t_s, \quad \Omega_K = \sqrt{\frac{GM_*}{R^3}}, \quad \mathrm{St}_0 \simeq \frac{\pi \rho_\bullet a}{2\Sigma_g}$$

- Ω_K：开普勒角频率；R：柱坐标半径
- Σ_g：气体面密度；下标 0 表示盘中平面
- 最右式要求 Epstein 阻力、等温高斯气体层与 $H_g = c_s/\Omega_K$
- $\mathrm{St} \ll 1$：紧耦合；$\mathrm{St} \sim 1$：阻力与轨道时间可比
- $\mathrm{St} \gg 1$：弱耦合；同一粒径在不同位置有不同 St

在盘动力学中，比较 St 比仅比较粒径a更有意义。

5 气体亚开普勒旋转：压强梯度的作用

PRESSURE SUPPORT AND HEADWIND

$$v_{g,\phi}^2 = v_K^2 + \frac{R}{\rho_g}\frac{dP}{dR}, \quad v_{g,\phi} \simeq v_K(1-\eta), \quad \eta = -\frac{1}{2}\frac{c_s^2}{v_K^2}\frac{d\ln P}{d\ln R}$$

- $v_K = R\Omega_K$；采用局部等温关系 $P = \rho_g c_s^2$
- 通常 $dP/dR < 0$，因此 $\eta > 0$：气体转速较低
- 开普勒运动的尘埃受到逆风，失去角动量
- 在压力峰内侧，dP/dR 可为正，漂移方向随之反转

6 径向漂移推导 I：局部稳态方程

RADIAL DRIFT: LOCAL FORCE BALANCE

$$0 = 2\Omega_K u_{d,\phi} - \frac{u_{d,R} - u_{g,R}}{t_s}, \qquad 0 = -\frac{1}{2}\Omega_K u_{d,R} - \frac{u_{d,\phi} - u_{g,\phi}}{t_s}$$

- u_ϕ 是相对当地开普勒转速的方位速度偏差；u_R 是径向速度
- $u_{g,\phi} = -\eta v_K$；局部展开采用开普勒剪切 $q = 3/2$
- 第一式：径向惯性力与阻力平衡
- 第二式：方位惯性力与阻力平衡
- 假设轴对称、稳态、并忽略尘埃反馈

7 径向漂移推导 II：消元得到速度

RADIAL DRIFT: SOLUTION

$$u_{d,R} = \frac{u_{g,R} - 2\eta v_K \mathrm{St}}{1 + \mathrm{St}^2}, \qquad u_{d,\phi} = \frac{-\eta v_K - \frac{1}{2}\mathrm{St}\, u_{g,R}}{1 + \mathrm{St}^2}$$

- 第二个平衡式给出 $u_{d,\phi} = u_{g,\phi} - \mathrm{St}\, u_{d,R}/2$
- 代入第一式，得到 $(1 + \mathrm{St}^2)\, u_{d,R} = u_{g,R} + 2\,\mathrm{St}\, u_{g,\phi}$
- 再使用 $u_{g,\phi} = -\eta v_K$，得到上面的径向速度
- 第一项描述气体径向输运，第二项描述压强驱动漂移
- 结果适用于给定背景气体，不能直接用于强反馈区域

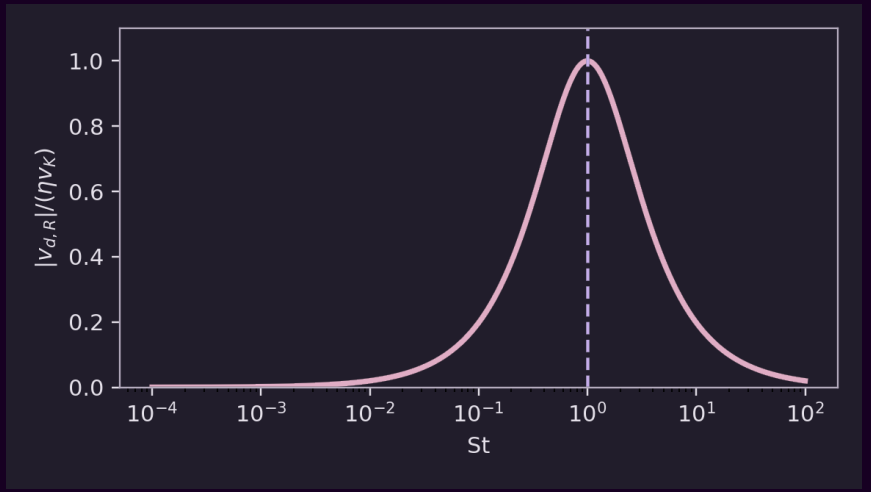
$\mathrm{St} \to 0$ 时 $u_{d,R} \to u_{g,R}$；紧耦合尘埃随气体径向流动。

8 为什么 St ≃ 1 漂移最快?

THE FASTEST DRIFTERS

$$u_{g,R} = 0: \qquad u_{d,R} = -\frac{2\eta v_K \mathrm{St}}{1+\mathrm{St}^2}$$

- St ≪ 1：速度差被drag迅速消除
- St ≫ 1：阻力太弱，角动量损失慢
- St = 1：径向速度大小达到 ηv_k
- 适用 $\eta > 0$、给定气体、无反馈



9 漂移时间

DRIFT TIME AND THE DRIFT BARRIER

$$t_{\rm drift} \equiv \frac{R}{|u_{d,R}|} = \frac{1+{\rm St}^2}{2\eta\,{\rm St}}\,\Omega_K^{-1}$$

- 假设 $u_{g,R} = 0$，且背景量在局部变化缓慢
- 示例：$M_* = 1\ M_\odot$，$R = 30$ AU，$H_g/R = 0.05$
- $d \ln P / d \ln R = -2.5 \rightarrow \eta = 3.125 \times 10^{-3}$
- $\Omega_K^{-1} \approx 26.2$ yr；St = 1 时 $t_{\rm drift} \approx 8.4 \times 10^3$ yr

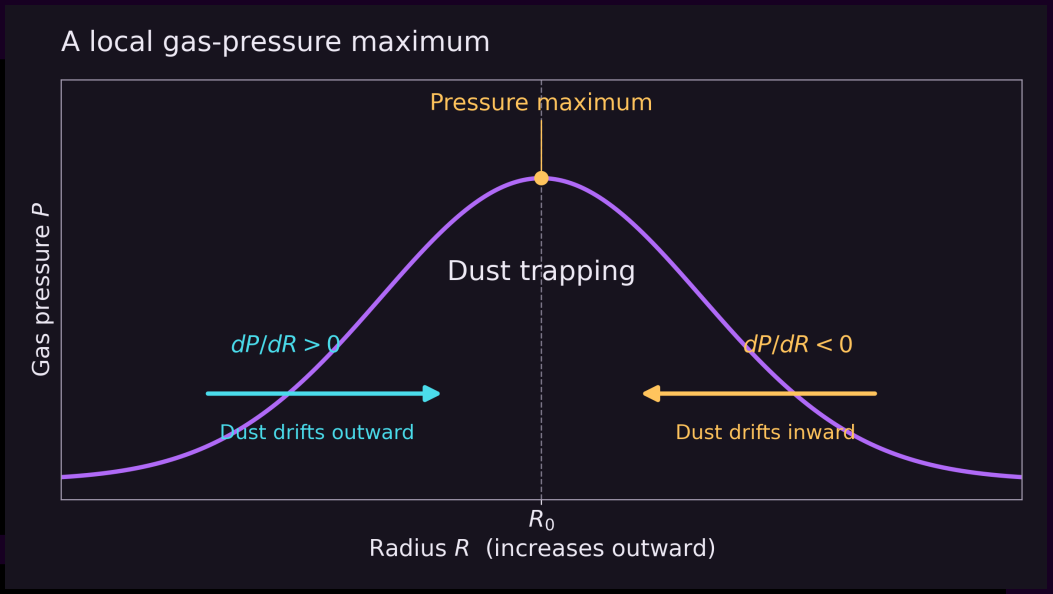
漂移可能比盘寿命短得多；增长与压力峰捕获必须与它竞争。

10 压力峰

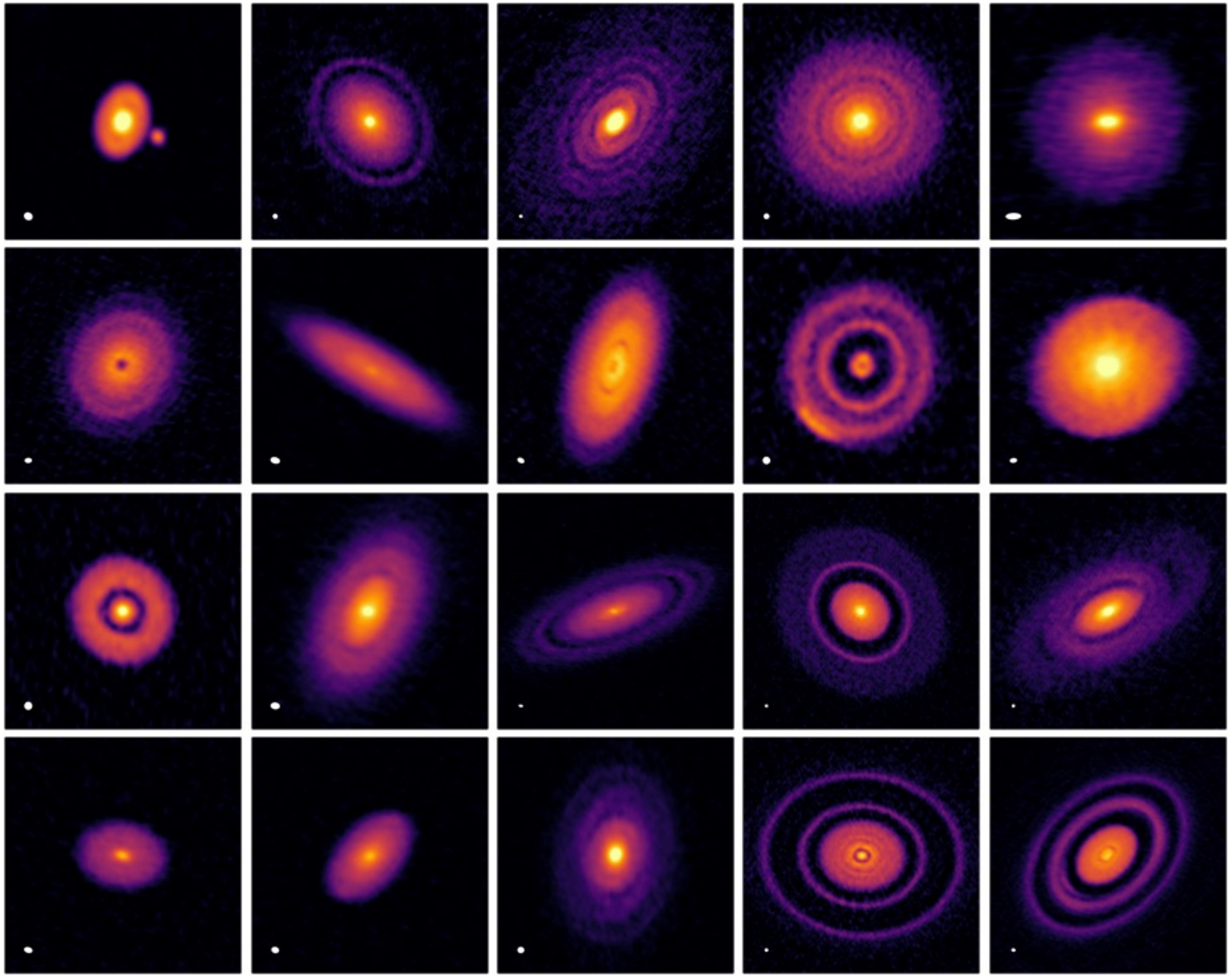
PRESSURE MAXIMA AND DUST TRAPPING

$$\eta < 0 \Rightarrow u_{d,R} > 0, \qquad \eta = 0 \Rightarrow u_{d,R} = 0, \qquad \eta > 0 \Rightarrow u_{d,R} < 0$$

- 压力峰内侧：dP/dR > 0，尘埃向外
- 压力峰外侧：dP/dR < 0，尘埃向内
- 两侧漂移都指向压力最大值
- 图与公式忽略气体径向流动
- 有气体吸积时，零漂移点可能偏移



行星、磁流体过程与其他盘结构均可产生压力峰。



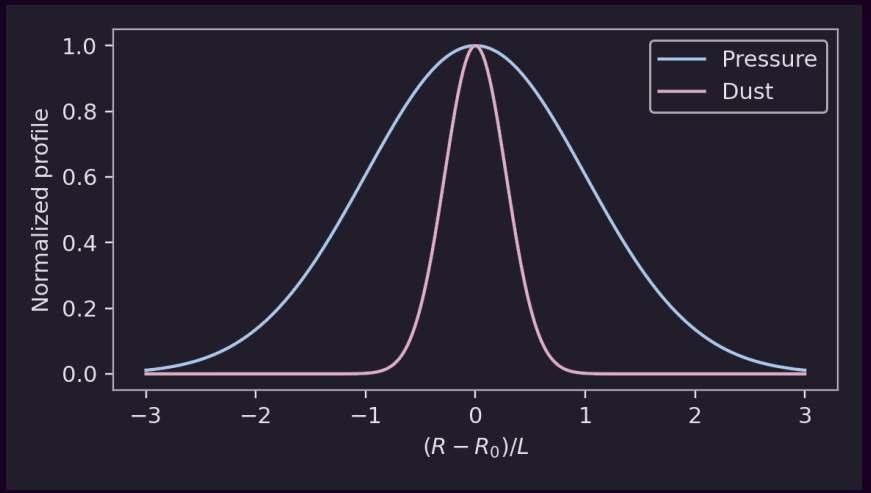
11 尘埃环宽度：捕获与扩散竞争

DUST RING WIDTH

$$\ln P \simeq \ln P_0 - \frac{x^2}{2L^2}, \quad v_{d,R} \simeq -\frac{\mathrm{St}\, c_s^2}{\Omega_K L^2} x, \quad w_c = L\sqrt{\frac{\delta_R}{\mathrm{St}}}$$

- $x = R - R_0$；L：局部压力峰宽度
- 假设 $\mathrm{St} \ll 1$，c_s、St、$D_{d,R}$ 恒定
- 零通量：$\partial_x \ln(\Sigma_d/\Sigma_g) = v_{d,R}/D_{d,R}$
- 得到尘气面密度比的高斯宽度 w_c
- 若 Σ_g 也为宽度 L 的高斯，尘埃宽度

  $w_d = L\,[1 + \mathrm{St}/\delta_R]^{-1/2}$



环宽约束 St/δ_R；单独测环宽不能唯一确定粒径或湍流。

12 湍流扩散

TURBULENT DIFFUSION

$$\mathbf{F}_{\mathrm{diff}} = -\rho_g D_d \nabla\left(\frac{\rho_d}{\rho_g}\right), \quad D_{g,R} = \delta_R c_s H_g, \quad D_{d,R} \simeq \frac{D_{g,R}}{1+\mathrm{St}^2}$$

$$\frac{\partial \rho_d}{\partial t} + \nabla\cdot(\rho_d \mathbf{v}_d + \mathbf{F}_{\mathrm{diff}}) = 0$$

$$\frac{\partial \rho_d}{\partial t} + \nabla\cdot(\rho_d \mathbf{v}_d) = \nabla\cdot\left[\rho_g D_d \nabla\left(\frac{\rho_d}{\rho_g}\right)\right]$$

- 无源汇时，F_{diff} 是尘埃相对于气体的扩散质量通量。
- ρ_d/ρ_g：尘气体积密度比；D_d：尘埃扩散系数。
- 扩散通过 $\nabla\cdot F_{diff}$ 进入连续性方程，不是加速度项。
- δ_R 是径向扩散参数，不必等于角动量输运参数 α。
- 垂直扩散可不同，记作 $D_{d,z} = \delta_z c_s H_g$。

闭合依赖湍流统计与方向；参考 Youdin & Lithwick (2007)。

13 尘埃输运方程

THE RADIAL ADVECTION–DIFFUSION EQUATION

$$\frac{\partial \Sigma_d}{\partial t} + \frac{1}{R}\frac{\partial}{\partial R}\left[R\left(\Sigma_d u_{d,R} - D_{d,R}\Sigma_g \frac{\partial}{\partial R}\frac{\Sigma_d}{\Sigma_g}\right)\right] = S_d$$

- Σ_d $u_{d,R}$：平流通量；第二项：相对于气体的扩散通量
- S_d：所选粒径组的碰撞转换、源或汇，单位为面密度/时间
- 采用轴对称、垂直积分描述；各量为有效平均量
- 对所有粒径求和时，质量守恒碰撞项相互抵消
- 初始分布、边界通量和气体演化共同决定最终结果

用于全球尘埃演化；需同时指定阻力、扩散和粒径演化模型。

14 垂直沉降

VERTICAL SETTLING

$$\ddot{z} + \frac{\dot{z}}{t_s} + \Omega_K^2 z = 0, \qquad \mathrm{St} \ll 1: \quad v_{d,z} \simeq -\Omega_K^2 t_s z$$

- 薄盘 $|z| \ll R$：垂直恒星引力为 $-\Omega_K^2 z$
- 假设 $v_{g,z} = 0$，忽略湍流与尘埃反馈
- 紧耦合极限：垂直加速度可忽略，达到终端速度
- 若 t_s 为常数：$z(t) = z_0 \exp(-\mathrm{St}\,\Omega_K\, t)$
- $t_{\mathrm{settle}} \simeq 1/(\mathrm{St}\,\Omega_K)$；大 St 时需保留垂直惯性

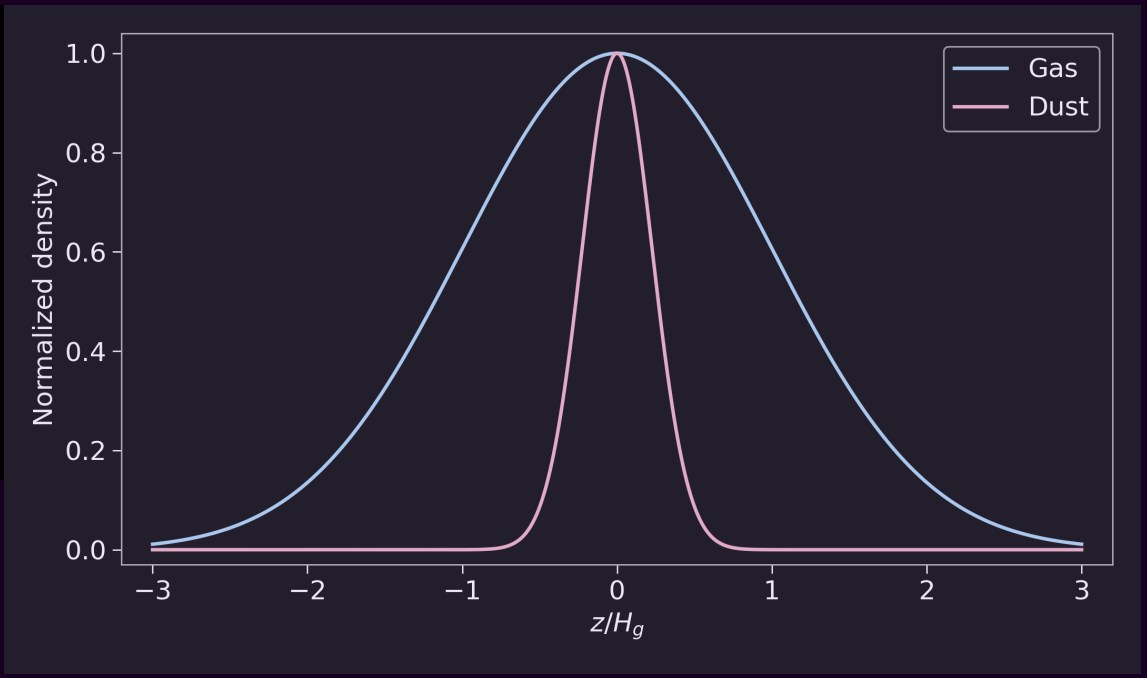
颗粒向盘中平面沉降；没有搅拌时，尘埃层会逐渐变薄。

15 沉降与扩散平衡：尘埃层厚度

SETTLING–DIFFUSION EQUILIBRIUM

$$\rho_d v_{d,z} - \rho_g D_{d,z} \partial_z (\rho_d/\rho_g) = 0 \quad \Rightarrow \quad \frac{H_d}{H_g} = \sqrt{\frac{\delta_z}{\delta_z + \mathrm{St}}}$$

- 设气体为高斯层，$H_g = c_s/\Omega_K$
- 使用小 St 的终端沉降速度
- 在局部将 St 和 $D_{d,z}$ 视为常数
- H_d 越小，盘中平面尘埃越集中



16 面密度比与中平面密度比

MIDPLANE DUST ENRICHMENT

$$Z \equiv \frac{\Sigma_d}{\Sigma_g}, \qquad \epsilon_0 \equiv \frac{\rho_{d,0}}{\rho_{g,0}} = Z\frac{H_g}{H_d}$$

- 两种组分都用高斯垂直分布
- Z 衡量固体总量；ε_0 衡量局部反馈
- Z = 0.01，St = 0.1，$\delta_z = 10^{-4}$
- 上式模型给出 $H_d/H_g \approx 0.032$
- 因而 $\varepsilon_0 \approx 0.32$，并非只有 0.01

